

HOTO

SewMate

Magic in minutes

HANDHELD SEWING MACHINE

Screen Interaction







Must have

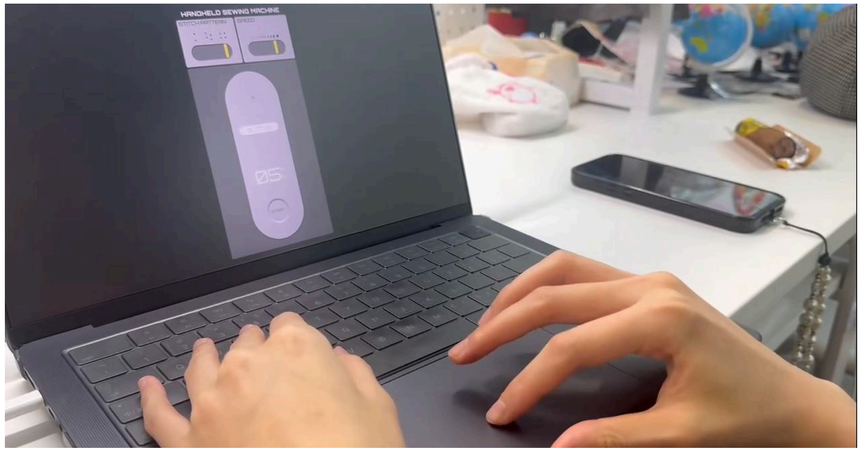
1. As a user, when I tap or drag the Stitch Pattern slider, I expect the needle bar moves along with it, and the preview, the status label (STITCHING / HEMMING / BUTTON) and the middle bar to update, so that I can confirm which stitch the machine will sew before I start.
2. As a user, when I move the Speed slider, I expect the real-time demo moves along with it.
3. As a user, when I pinch or spread two fingers on the stitch preview, I expect the stitch in the preview and in the middle bar to get smaller or bigger, so that I can adjust stitch size to fit my fabric.
4. As a user, when I press and hold the START button for one second, I expect the interface to turn on.
5. As a user, when I press and hold START again while the machine is on, I expect the display to turn off.
6. As a user, when I touch the Stitch Pattern slider while the machine is start, I expect nothing to change, for safety.

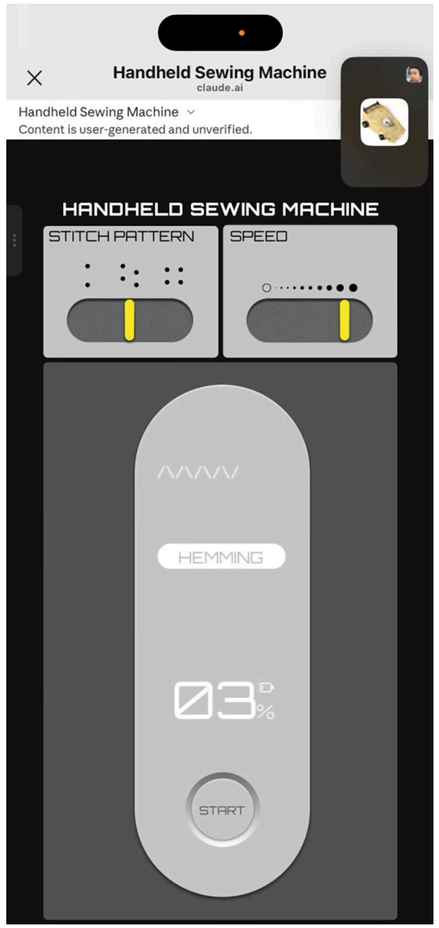
Nice to have

1. As a user, when I touch the Stitch Pattern slider while the machine is start, I expect nothing to change, for safety.
2. As a user, when I start the machine for a while, the battery goes down.

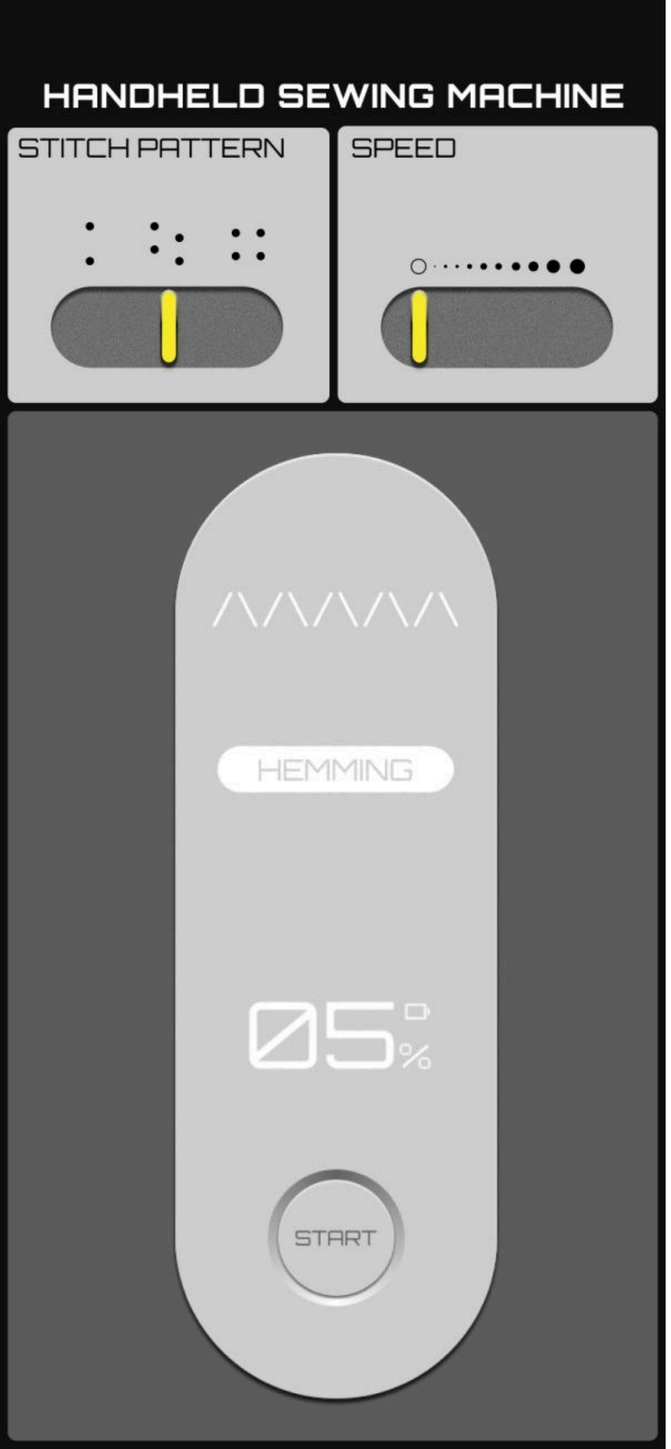
Dropped

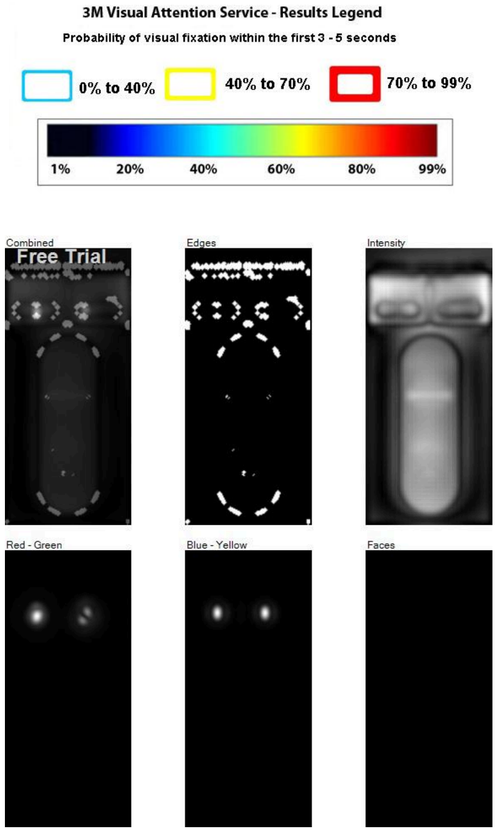
1. Tension lever reaction
2. Presser foot lever reaction
3. Real cloth stitching animation

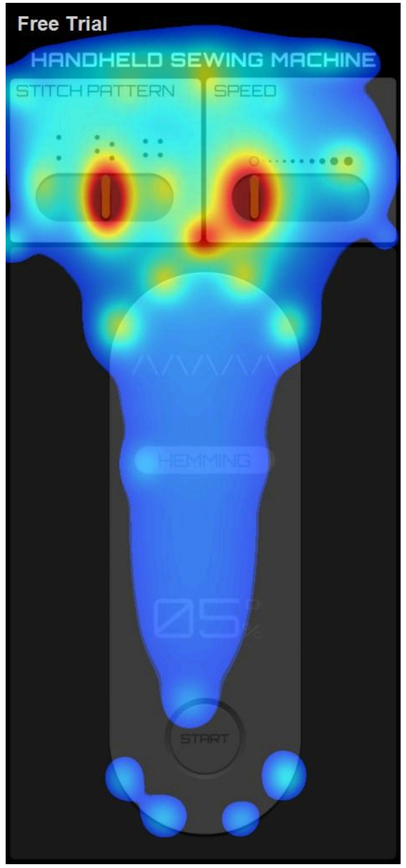

HANDHELD SEWING MACHINE
STITCH PATTERN
SPEED
HEMMING
START

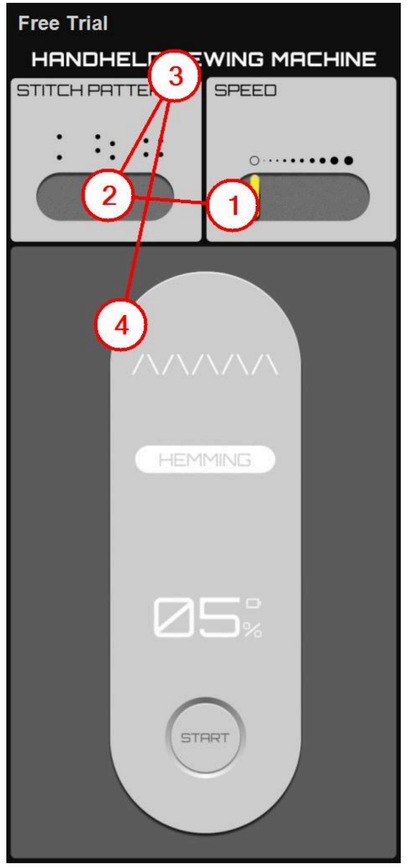

Handheld Sewing Machine
claude.ai
Handheld Sewing Machine
Content is user-generated and unverified.
HANDHELD SEWING MACHINE
STITCH PATTERN
SPEED
HEMMING
START




HANDHELD SEWING MACHINE
STITCH PATTERN
SPEED
HEMMING
START


3M Visual Attention Service - Results Legend
Probability of visual fixation within the first 3 - 5 seconds
0% to 40%
40% to 70%
70% to 99%
1%
20%
40%
60%
80%
99%
Combined
Free Trial
Edges
Intensity
Red - Green
Blue - Yellow
Faces


Free Trial
HANDHELD SEWING MACHINE
STITCH PATTERN
SPEED
HEMMING
START


Free Trial
STITCH PATTERN
SPEED
1
2
3
4
HEMMING
START

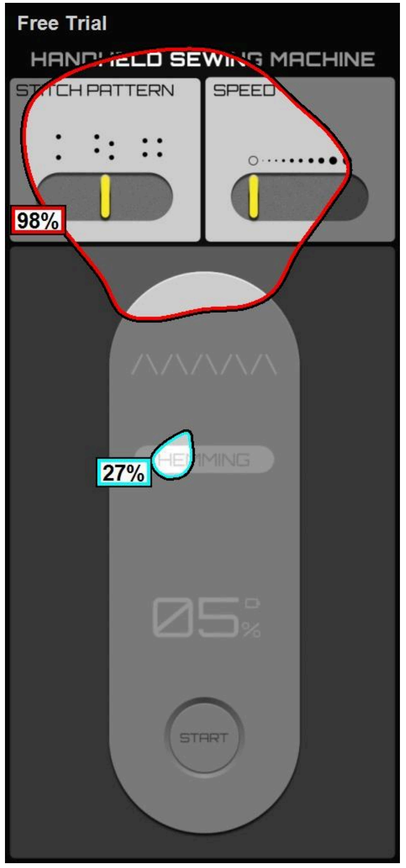

Free Trial
STITCH PATTERN
SPEED
98%
27%
HEMMING
START

Table A — Functionality

Outcome 4 unaided · 3 hesitation · 2 with help · 1 failed Severity 4 blocker · 3 major · 2 minor · 1 cosmetic

Task	Step	Priority	P#	Outcome	Severity	What happened
T1 Turn the machine on	S1 Find the power control	Must	Some	3 — hesitation	2 Minor	START not obvious at first
	S2 Hold START for 1 second	Must	Some	3 — hesitation	3 Major	Several tries to find hold vs tap
T2 Zigzag at medium speed	S1–S3 Select zigzag	Must	All	4 — unaided	—	No issues reported
	S4–S5 Set a medium speed	Must	All	4 — unaided	—	No issues reported
	S6–S7 Start it, see it sewing	Must	Some	3 — hesitation	2 Minor	Blinking warning hard to read
T3 Make the stitches bigger	S1–S2 Find and pinch the preview	Must	All	1 — failed	4 Blocker	No one tried to zoom in/out on the stitch pattern icon

Table B — Emotion

Score +2 delight · +1 positive · 0 neutral · −1 friction · −2 frustration Every score needs what you saw and their exact words

Element	P#	Score	Observed	Verbatim
“Select a stitch speed” warning	Some	−1	Blinking text was hard to read	It flashes too fast to see clearly.
Stitch pattern slider	Some	0	Worked as expected, no comment	Why is it inconsistent with the direction of the real-time demo?
Speed slider + live preview	Some	+1	Worked as expected, Someone didn’t understand why speed adjustment was necessary. I explained that the feed dogs in a handheld sewing machine aren’t powerful enough to move the fabric forward on their own, so users need to pull the fabric with their other hand. In this case, the sewing speed determines the stitch density.	Make sense
Pinch to resize stitches	All	-2	Nobody found it alone; all surprised once shown	Oh~~~ I see, interesting

Three findings

1

Pinch to resize is hidden

SAW

No one found it alone; all were surprised once shown

MEANS

A gesture with no cue doesn’t exist for users

DO

Show a pinch hint on the preview at first power-on

2

START is unclear

SAW

Several tries to tell hold (on/off) from tap (start)

MEANS

One button, two hidden actions

DO

Bolder START + a ring that fills while holding

3

Warning is hard to read

SAW

The blinking two-line text was hard to read

MEANS

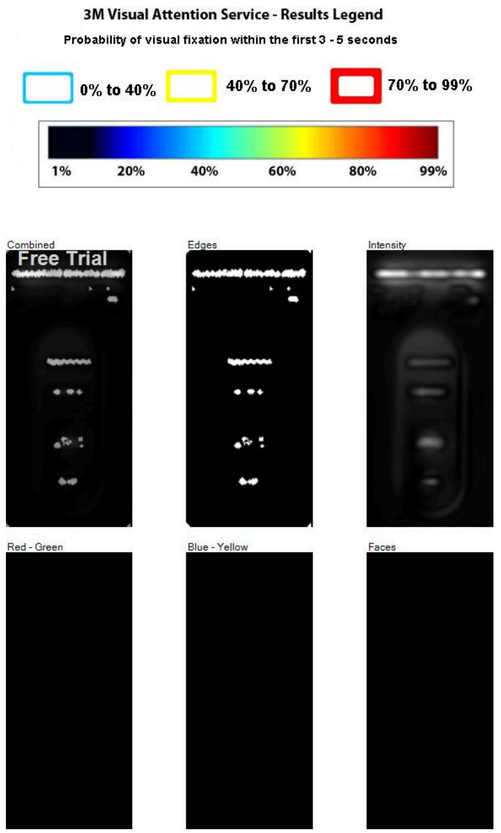
It alarms but doesn’t point to the fix

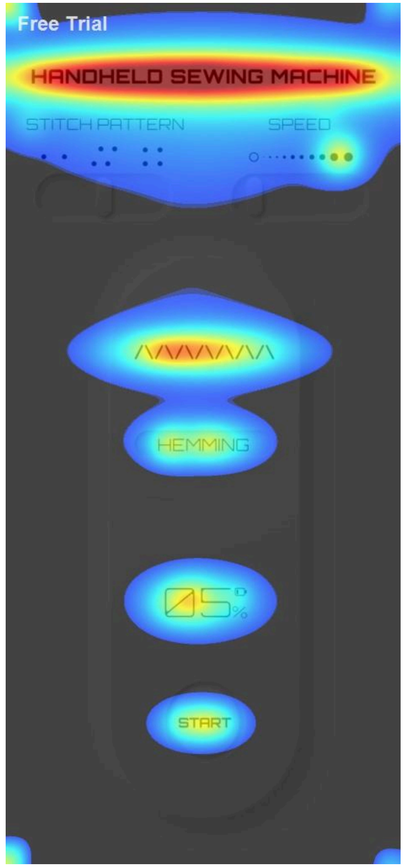
DO

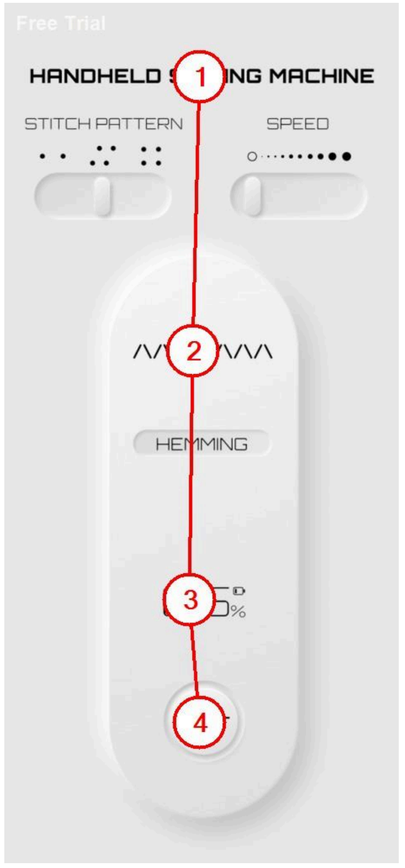
Steady “SET SPEED FIRST”, speed slider pulses

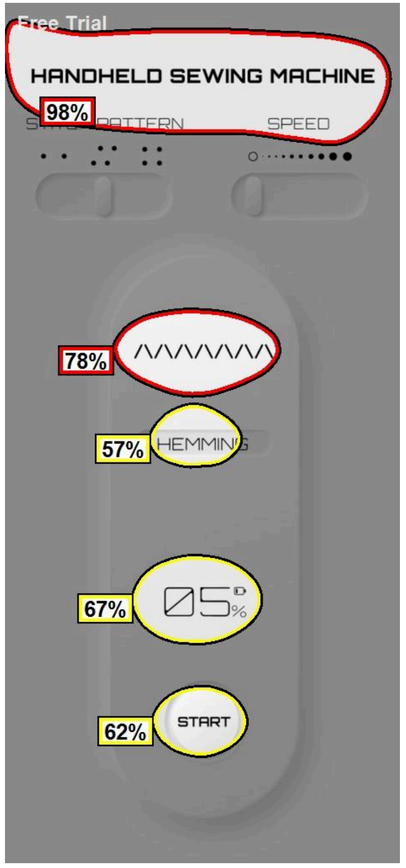
HANDHELD SEWING MACHINE













B104 Linear Slide Potentiometer



SMD Tactile Push Button Switch



ESP32-S3 3.49-inch Touchscreen Development Board



Rotary Coded DIP Switch

